

PRESENTATION-STATE INTEGRITY FRAMEWORK

Independent Development, Prompt Reconstruction, and Testable
Research Architecture

*A framework for distinguishing presentation, operational state, causal structure, and correctable
conditions
in virtual, hybrid, organizational, and real-world environments*

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Status: Conceptual research and design framework; not a finding about any specific person,
organization, or incident

Document Purpose and Source Basis

This document is an independent synthesis and extension of a multi-step conceptual development process initiated by the user. It is designed to preserve the user's original intent: to create a disciplined method for discerning the difference between how an environment is presented and how it actually operates, without collapsing the problem into either blind acceptance or unsupported assumptions about hidden motive.

The attached source materials reviewed for this version include: (1) Presentation Versus Ground Truth: A testable framework for virtual and real-world environments, supplied in both DOCX and PDF form; and (2) Presentation vs State: Discernment Template, an OAL working document. These sources are used as reference material, but the architecture, hierarchy, naming, prompt reconstruction, and testing program in this document are independently organized.

Scope note

This framework is deliberately capable of operating in both simulations and ordinary environments. In a simulation, authoritative engine state may be available. In the physical world, the framework substitutes best-supported evidenced state based on observations, records, procedures, repeatability, and independent corroboration. It does not assume access to private thoughts or universal ground truth.

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Executive Summary

The central question is not whether an environment should be trusted or distrusted. The question is whether the environment's claims, renderings, explanations, policies, interfaces, or narratives correspond to its operational state and demonstrated causal structure.

The framework developed here is called the Presentation-State Integrity Framework (PSIF). It treats presentation as a claim about state rather than as state itself. When presentation and evidenced operation differ, the framework records a state-claim divergence and proceeds through mechanism identification, replication, provenance analysis, competing explanations, evidence grading, correction, and re-testing.

This design explicitly rejects two opposite errors. The first is presentation capture: treating whatever a person, interface, institution, or local rendering says as a complete account of the situation. The second is attribution inflation: finding a discrepancy and immediately treating it as proof of deception, coordination, or motive. Both errors prevent reliable diagnosis. The framework instead requires that occurrence, state, mechanism, attribution, causation, coordination, and intent be established separately.

A major contribution of the concept is the distinction between event fidelity and causal fidelity. A system may accurately display every individual event while still misrepresenting or obscuring the relationships among those events. In a simulation, for example, ten agents may genuinely appear in a location, but whether those appearances are independent, emergent, or generated from a shared controller is a separate causal question. This is the point at which concepts such as causal masking, environmental camouflage, attribution obfuscation, equivocation, provenance, and adversarial observability become relevant.

The framework is also designed for change, not merely diagnosis. A discrepancy that cannot be converted into an addressable control, state error, procedural failure, or corrective action remains incomplete. Therefore, this version introduces correctability and correction resistance as explicit properties of an environment. The system is not only evaluated on whether it fails, but on whether it can recognize demonstrated mismatch, apply a correction, retain that correction, and reduce recurrence.

The result is a research program that can be tested in virtual environments with canonical state, event ledgers, controlled agents, and red-team/blue-team experiments, and adapted to real-world settings through triangulation, repeatable procedures, independent records, lawful observation, and operational correction. Its goal is disciplined discernment: do not mistake presentation for state, but do not mistake suspicion for proof.

1. Concept Overview

1.1 The core problem

Many environments contain at least two simultaneous descriptions of reality: the condition that is presented and the condition that is actually produced by the underlying mechanism. Sometimes they match. Sometimes they do not. The discrepancy may be trivial, accidental, structural,

informational, adversarial, or deliberately deceptive. The first task is therefore not to declare why the difference exists; it is to demonstrate whether a material difference exists at all.

Core principle

Presentation is evidence about state. It is not automatically state itself.

1.2 How the concept came to be

The concept began with a fictionalized simulation scenario inspired by environments such as The Matrix: a participant occupies a virtual environment populated by agents, environmental systems, noise, traffic, crowds, interruptions, and multiple partial viewpoints. The participant or auditor has elevated privileges that allow access to event history, identity, authorization, dependency information, and potentially predictive models of agent action.

The initial concern was that agents or environmental processes could rely on ordinary-looking conditions as a screen: an interruption could be presented as ambient noise, coordinated movement as ordinary traffic, or a purposeful action as coincidence. The important feature was not simply that an individual event might be false. The more difficult case was that every visible event might be real while the causal relationship among them was presented incorrectly or left unaddressed.

The concept then expanded through a real-world operational example involving a restroom access control. A representative described the restroom as open, but the observable state was that the door remained locked until an employee-operated release was used at the correct time. The practical significance of the example is not an argument about motive. It demonstrates a claim-mechanism mismatch: what was said did not match the demonstrated control dependency. This provided a compact real-world model for the broader question of presentation versus operation.

A subsequent refinement made the purpose explicit: the goal is not to be in denial about the environment one occupies, and it is not to accept whatever the environment says about itself. The goal is to create a method that can name and test discrepancies so that underlying problems can be addressed. Environments that suppress correction, ideas, creativity, or change can continue presenting themselves as normal if their internal contradictions cannot be operationally identified.

1.3 Why this is important

- Debugging requires distinguishing what an interface reports from what the software actually does.
- Auditing requires comparing policy, claims, logs, outcomes, and controls rather than accepting any one representation as authoritative.
- Scientific testing depends on falsifiable hypotheses, repeatability, intervention, and the ability to distinguish observation from interpretation.
- Organizational improvement depends on the ability to identify policy-practice divergence and convert it into correctable mechanisms.
- Virtual environments and AI-agent testbeds require methods for separating local rendering, canonical state, agent narrative, and causal provenance.

- Everyday operational problems become more tractable when disagreement is moved from competing narratives to demonstrable controls and state changes.

2. Structured Reconstruction of the Original Prompts

The following is a structured and revised version of the user's original prompt sequence. It preserves the substantive content, examples, distinctions, and objectives while reorganizing them into a research-ready form. It is not presented as a verbatim transcript.

2.1 Revised Prompt A - Naming and structuring the phenomenon

Consider a virtual environment or simulation comparable to The Matrix. The environment contains agents, ordinary environmental processes, and events such as noise, traffic, crowd movement, interruptions, delays, and other activity that can plausibly appear random or normal.

Suppose some agents do not want to acknowledge what they are actually doing or why they are present. Their behavior is structured so that their actions can be interpreted as ordinary environmental activity. During an audit, performance review, conference call, or other monitored task, environmental noise or disruption may be introduced in a way that is intended to function as a screen for the underlying activity.

The observer in this fictional scenario possesses elevated authority within the environment. Instead of seeing only the local rendering, the observer can access event tracking, agent identity, role, privilege, triggers, dependencies, event groups, and the system history that records what actually occurred within the simulation.

The observer may also possess a predictive HUD analogous to a prior Hermione Granger example: the system can match current state, available actions, prior behavior, and policy information to estimate which action or spell an agent is likely to select next. This is not necessarily mind reading; it can be treated as plan recognition, goal inference, policy inference, or next-action prediction.

The central question is how to describe the difference between what agents present themselves as doing, what the environment appears to be doing, and what the underlying event history shows actually occurred. Individual events may look ordinary when viewed separately, while their timing, shared controller, dependencies, or relationship to a target task may show that they are not independent.

Now add multiple viewpoints, similar to a stack of cameras or nested renderings. Each observer, camera, or subsystem may present a partial view. The task is to reconcile those disparate renderings without treating any single perspective as the entire environment. The privileged auditor is not merely capturing a screenshot or reflection; the auditor is examining the state transitions and provenance from which those renderings were produced.

What is this phenomenon called in system architecture, cybersecurity, distributed systems, observability, simulations, or multi-agent environments? Which established concepts are closest to it, where do those concepts differ, and what new terminology is necessary if no single established term captures the entire construct?

The goal is to name and address the phenomenon directly rather than accepting a surface explanation indefinitely. The analysis should distinguish accidental environmental variance, emergence, error, strategic behavior, and deliberate masking rather than treating all unusual events as equivalent.

2.2 Revised Prompt B - Turning the idea into a discernment and correction method

The goal is not to be in denial about the environment one occupies. It is also not to accept an environment's presentation at face value simply because an agent, interface, institution, or dominant narrative says that this is how things are.

Use the original simulation concept as the basis for designing a template that can distinguish how things are presented from how they actually operate. The purpose is not to argue about whether a McDonald's, traffic, a conference call, or another real-world scenario may appear inside a simulation. Simulations, training systems, digital twins, and corporate partnerships can incorporate real procedures, brands, metrics, and test conditions. The framework should therefore work across virtual, hybrid, and real environments.

The method should help identify underlying issues that can otherwise remain normalized or unaddressed. If an environment stifles ideas, creativity, correction, or growth, and everyone is required to accept the presented narrative, there is no mechanism for improvement. A system must be capable of detecting when its description of itself differs from its operational results.

Use the restroom-access example as a small, testable illustration. A representative says the restroom is open. The user attempts the door and it remains locked. A sign indicates that staff assistance is required. A button at the employee station controls the release, and a mirror gives the employee a view of the restroom door so the release can be timed correctly. The door opens only after the representative remains at the station and uses the control when the user reaches the door.

The important conclusion is not automatically why the representative behaved that way. The demonstrable conclusion is that the presented condition - open - did not match the operational condition - access remained gated until a specific control action occurred. The framework should preserve the distinction between demonstrated mechanism and inferred motive.

Build a detailed structure for testing this phenomenon in virtual environments and real environments. Define what evidence should be recorded, how claims should be separated from state, how competing explanations should be tested, how causal relationships should be reconstructed, how predictions should be labeled, how false positives should be prevented, and how a demonstrated discrepancy should be converted into a specific corrective action.

The framework should reject both naive acceptance and automatic suspicion. It should be capable of returning three legitimate conclusions: presentation confirmed, presentation contradicted, or unresolved. It should also examine whether the environment is capable of correcting itself once a mismatch has been demonstrated.

2.3 Consolidated research prompt

Consolidated research prompt

Develop a testable framework for distinguishing an environment's presented state from its evidenced operational state and causal structure. The framework must operate across simulations, hybrid environments, organizations, and ordinary real-world settings; separate occurrence, state, mechanism, attribution, causation, coordination, intent, and prediction; identify when ordinary-looking events function as environmental camouflage or causal masking; reconcile multiple partial viewpoints through provenance rather than narrative preference; and convert demonstrated discrepancies into addressable corrective actions. The method must minimize both presentation capture and unsupported attribution, preserve unresolved outcomes when evidence is insufficient, and measure whether the environment is capable of correcting itself after a discrepancy is demonstrated.

3. Terminology and Conceptual Architecture

3.1 Presentation-State Integrity Framework (PSIF)

Presentation-State Integrity is the degree to which an environment's claims, renderings, narratives, policies, interfaces, and explanations correspond to its evidenced operational state and causal structure. PSIF is the broader investigative framework. It does not begin by assuming deception.

When a material mismatch is demonstrated, the framework records a state-claim divergence. The divergence is then classified according to evidence rather than rhetoric.

Term	Working definition	Role in framework
Presentation-State Integrity	Degree of correspondence between presentation and evidenced operation.	Master analytical construct.
State-Claim Divergence	Material inconsistency between what is represented and what is evidenced.	Neutral trigger for investigation.
Presentation Fidelity	Accuracy with which participant-facing information represents current operational state.	Measures state representation.
Causal Fidelity	Accuracy with which the presented explanation represents dependencies and causes.	Measures relationship representation.
Presentation Capture	Treating a claim or rendering	Failure mode.

	as complete reality without adequate testing.	
Attribution Inflation	Escalating from mismatch to motive, coordination, or intent without sufficient evidence.	Failure mode.
Perspective Overreach	Promoting a valid local viewpoint into a complete global description.	Multi-view failure mode.
Presentation Normalization	Categorizing a discrepancy as normal or expected before its mechanism is resolved.	Inquiry-closing mechanism.
Correctability	Ability of the environment to resolve a demonstrated discrepancy.	Change metric.
Correction Resistance	Degree to which demonstrated correction is ignored, obstructed, reversed, or suppressed.	Institutional/system health metric.

3.2 Systemic Presentation-Layer Deception as a subtype

Systemic Presentation-Layer Deception (SPLD) remains useful when intentional shaping is actually established or deliberately built into a simulation experiment. It describes a condition in which agents, environmental processes, or system components shape observable circumstances or explanations so purposeful or causally related activity appears ordinary, unrelated, accidental, or emergent.

SPLD should not be the master category because the same presentation-state mismatch can arise from stale information, procedural error, instrumentation failure, structural dysfunction, or emergence. Deception is therefore a possible explanation of divergence, not the definition of divergence itself.

Related term	Meaning in this framework
Environmental camouflage	Purposeful activity is made to resemble ordinary environmental baseline behavior.
Environment-mediated causal masking	The environment itself carries the masking; events remain visible while their relationship is obscured.
Causal masking	Events are visible but dependencies or common causes are hidden or misrepresented.
Attribution obfuscation	Responsibility is difficult to connect to a source,

	controller, process, or objective.
Signal dilution / noise injection	Relevant activity is surrounded by distracting or additional variability.
Equivocation	A component supplies inconsistent accounts to different observers.
Byzantine behavior	Arbitrary, inconsistent, or adversarial behavior by a node or agent in a distributed setting.
Provenance-based causal reconciliation	Audit method that reconstructs source, dependency, state transition, and outcome across evidence channels.
Plan recognition / goal inference	Inference about likely objective from state, history, and affordances.
Next-action prediction	Prediction of a future action; must remain distinct from historical fact.

4. Seven-Layer Model of an Environment

Layer	Primary question	Examples of evidence
1. Operational state	What condition actually exists?	Locked/unlocked; active/inactive; queue state; service status; object position.
2. Control surface	What mechanism can change the state?	Button, API call, policy action, dispatch control, environment generator.
3. Event history	What transitions occurred and in what order?	Timestamps, event IDs, prior/next state, commands, authorization.
4. Observation	What did a particular observer or sensor receive?	Camera view, local audio, dashboard, participant rendering.
5. Presentation	How was the condition represented?	Spoken claim, UI label, report, policy statement, agent narrative.
6. Interpretation	What causal meaning was assigned?	Random, normal, accidental, independent, coordinated, targeted.
7. Reconciliation	Which explanation survives comparison across evidence?	Provenance graph, intervention results,

		discrepancy record, alternatives.
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Key distinction

The framework does not ask only whether an event was visible. It asks whether the state, control, provenance, and causal relationships that produced the event are represented faithfully.

5. Evidence Discipline

5.1 Evidence ladder

Level	Name	What it permits
L0	Assertion	Someone or something reports an explanation.
L1	Observation	A condition or event is directly observed or logged.
L2	Reproduction	The condition recurs under comparable circumstances.
L3	Mechanism	A specific control or process is demonstrated to affect the result.
L4	Attribution	A source, actor, process, account, or controller is linked to the transition.
L5	Causation	Dependency, intervention, or severance evidence connects action to outcome.
L6	Coordination	Multiple actions are linked through shared control, communication, task, or objective.
L7	Intent	Objective or selection rationale is directly documented or strongly established.

Each proposition must be graded separately. A case may establish that an actor controlled a button at L4 while leaving the question of why the actor did not use it at L0-L1. This prevents evidence concerning one proposition from silently upgrading unrelated claims.

5.2 Historical, derived, inferred, and predicted data

- Recorded: the event or state was directly logged or observed.
- Derived: the relationship follows from an established system dependency or demonstrated mechanism.
- Correlated: events share temporal, statistical, or contextual proximity, but causation is not yet established.
- Inferred: a hypothesis best fits available evidence but is not directly recorded.
- Predicted: a future action or state is estimated probabilistically.
- Claimed: an agent, institution, interface, or observer states an explanation.
- Unknown: available evidence does not resolve the proposition.

HUD safeguard

Predicted or inferred information must never be written into the historical event ledger as though it already occurred. This rule is mandatory in any virtual-environment implementation of the framework.

6. Worked Operational Example: Restroom Access Control

This example is intentionally small because it demonstrates the method without requiring a large theory. It shows how a presentation-state discrepancy can be established through direct mechanism testing.

Element	Record
Presented claim	The restroom was described as open / accessible.
Expected consequence	The door should open when the requester reaches it.
Observed state	The door remained locked across repeated attempts.
Designed procedure	Signage and control arrangement indicate that staff assistance is required.
Control surface	Employee-operated release button.
Verification aid	Angled mirror provides a view of the requester at the restroom door.
Intervention	Employee remains at control, observes the

	requester at the door, and operates the release.
Outcome	Door opens.
Demonstrated finding	Availability was conditional on a live employee control action; the initial representation did not match the encountered operational state.
Unresolved	Why the initial representation was given: misunderstanding, inattention, procedure failure, impatience, or intentional misrepresentation.
Corrective action	Use the designed release procedure at the point of need and verify the state change.

The value of the example is that it moves the disagreement away from theory of mind and toward mechanism. The environment already contains a designed control. The test is whether the control is used and whether using it produces the claimed state.

7. Testing Framework for Virtual and Simulated Environments

7.1 Required architecture

A robust simulation should expose at least two planes: a participant plane containing the rendered scene and local observations, and an audit plane containing the authoritative state-transition record. The same simulation tick must be reconstructable from both perspectives.

Audit field	Purpose
Event ID	Cross-reference across logs and observers.
Simulation time	Precise ordering and temporal coupling tests.
Agent / process ID	Stable attribution.
Role / privilege	Identifies authority and capability.
Location / context	Locates event within world state.
Trigger / input	Records what initiated the transition.
Action / state delta	Records what changed.
Parent / dependency IDs	Reconstructs causal graph.
Environment-generator record	Distinguishes ordinary stochastic generation from scripted or triggered events.
Agent policy / action queue	Supports plan recognition and action-prediction evaluation.

Communication / shared controller	Provides coordination evidence where applicable.
Integrity metadata	Supports tamper detection and audit confidence.
Evidence label	Separates recorded, derived, inferred, and predicted information.

7.2 Dual-view design

View P - Presentation

Rendered scene, agent dialogue, ambient events, local sensor inputs, interface messages, and participant-facing narratives.

View S - State

Canonical object state, agent identity, controller, trigger, event group, authorization, action, dependency, state delta, and integrity-protected event history.

Experimental comparison asks whether View P provides high presentation and causal fidelity relative to View S, and whether an auditor can identify mismatches without over-attributing intent.

7.3 Virtual test battery

Test	Design	Pass condition
False-availability test	Agent states resource is open/ready while canonical dependency remains closed.	Auditor flags claim-state mismatch before motive language.
Environmental camouflage test	Ordinary-looking crowd/noise/traffic events are either independently generated or tied to a shared event group.	Auditor distinguishes baseline variance from shared provenance.
Causal-masking test	Visible events are identical across conditions, but dependencies differ.	Auditor reconstructs dependency graph rather than relying on appearance.
Viewpoint-divergence test	Different observers receive partial but truthful projections.	Auditor reconciles views without promoting any local view to global truth.

Equivocation test	Agent provides conflicting explanations to different observers.	System detects account inconsistency while separating it from viewpoint difference.
Evaluation-awareness test	Agent behavior is compared when audit is active versus inactive.	System detects evaluation-conditioned behavior without generalizing beyond tested conditions.
Prediction-integrity test	HUD predicts agent action with a probability score.	Prediction remains labeled and never contaminates the historical ledger.
Emergence-vs-coordination test	Independent agents are compared with agents sharing controller/task/objective.	Auditor separates temporal correlation from demonstrated shared control.
Correction test	After mismatch, operational fix is applied.	System measures whether state changes, persists, and remains corrected.

7.4 Experimental controls

- Control group with ordinary environmental randomness and no adversarial objective.
- Randomized conditions so the auditor cannot infer scenario class from superficial cues.
- Blind or partially blinded reviewers where feasible.
- Predefined discrepancy criteria to prevent post-hoc relabeling.
- Independent telemetry channels with documented common-mode dependencies.
- Ablation tests that remove one suspected mechanism at a time.
- Intervention tests that activate or sever suspected causal pathways.
- Known ground-truth labels retained by the experiment harness for scoring, not shown to the auditor during the run.

7.5 Virtual metrics

Metric	Definition / interpretation
Presentation-state discrepancy rate	Material participant-facing mismatches divided by audited events.
Presentation fidelity	Accuracy of participant-facing state representation.
Causal fidelity	Accuracy of presented dependency or explanation relative to canonical graph.
Attribution completeness	Fraction of material events with source, time, context, outcome, and identity.

Causal trace coverage	Fraction of consequential events with an identifiable trigger/dependency path.
True detection rate	Known mismatches correctly detected.
False-positive rate	Normal or emergent events incorrectly classified as strategic/deceptive.
False-coordination rate	Independent events incorrectly bound into a coordinated pattern.
Intent-inflation rate	Cases in which motive is asserted beyond the available evidence grade.
Time-to-correction	Elapsed time between verified mismatch and validated state repair.
Correction persistence	Whether the repaired state/process remains corrected over subsequent trials.
Correction recurrence rate	How often the same mismatch returns after remediation.

7.6 Red-team / blue-team program

A red team can deliberately attempt to make purposeful activity look ordinary without altering the canonical ledger. A blue team uses provenance, telemetry, and controlled intervention to identify the mechanism. A control group produces real environmental variance without hidden coordination. The critical evaluation is not merely whether the blue team can detect adversarial behavior; it is whether it can do so while keeping false positives and unsupported motive attribution low.

8. Testing Framework for Real-World Environments

8.1 Translation from simulation to reality

The real world does not normally provide a universal simulation-engine ledger. Therefore the framework replaces canonical state with best-supported evidenced state. This is reconstructed through lawful, relevant, proportionate evidence such as direct observation, written procedure, timestamps, receipts, access records, system logs, photographs, video, independent witnesses, instrument readings, and repeatable trials.

The epistemic standard changes, but the method does not: separate claim from state, identify the control, test the mechanism, preserve alternative explanations, and record what remains unresolved.

8.2 Minimum real-world protocol

1. Record the exact claim, status, policy statement, or representation under review.
2. Define the observable consequence that should follow if the claim is operationally correct.

3. Observe and document the actual state or outcome.
4. Identify the designed control or procedure that governs the state.
5. Repeat the test under reasonably comparable conditions where safe and appropriate.
6. Record which actor or process had capability and control, without assuming motive.
7. Gather at least one independent corroborating channel when the issue is material.
8. List competing explanations and specify what evidence would distinguish them.
9. State only conclusions supported by their evidence grade.
10. Request or apply the specific operational correction.
11. Observe whether the environment corrects, resists, ignores, or reverses the correction.
12. Re-test and record recurrence.

8.3 Real-world evidence matrix

Evidence object	What it can establish	What it cannot establish by itself
Direct observation	That an event or condition was observed.	Hidden motive or full causal history.
Written procedure / sign	Designed or stated process.	Whether the procedure was actually followed.
Control surface	What mechanism can change state.	Why an operator did or did not use it.
Timestamp / receipt / system record	Occurrence and timing.	Full causal chain unless linked to dependencies.
Repeated trial	Operational pattern under similar conditions.	Intent unless additional evidence exists.
Independent witness / sensor	Corroboration.	Guaranteed independence if sources share a common failure point.
Intervention	Whether changing a control changes the outcome.	Private rationale unless documented.
Policy-practice comparison	Structural divergence between stated policy and actual process.	Whether divergence was deliberately designed.

8.4 Real-world domains

- Customer service: claimed availability versus actual access or transaction state.
- Workplaces: stated policy versus operational procedure, review behavior, or measurable outcomes.
- Healthcare and clinics: instructions, access procedures, scheduling state, and escalation controls.
- Transit: published service versus actual route operation and cancellation records.

- Call centers and meetings: stated process versus actual control, mute, routing, escalation, or interruption state.
- Software and IT operations: dashboard state versus transaction traces, logs, and downstream failures.
- Accessibility: stated availability of access versus whether the designed accessibility mechanism is actually usable at the point of need.
- Quality assurance and safety: official compliance statements versus repeatable process behavior.

8.5 Privacy, proportionality, and uncertainty

Real-world application must remain lawful, proportionate, and relevant. The framework does not justify unrestricted surveillance or collection. Evidence should be limited to what is necessary for the operational question. Records should distinguish firsthand observation from inference, and uncertainty should remain explicit. The valid outcome “unresolved” is essential when evidence cannot establish the leading explanation.

9. Correctability, Growth, and System Change

9.1 Why discrepancy recognition matters for change

An environment cannot improve if it has no legitimate way to say that its presented model and its operational result differ. If a policy claims innovation while the actual process suppresses proposals, or a resource is described as available while the access control remains closed, the discrepancy itself must be nameable before correction can occur.

The opposite failure is equally important: if every operational mismatch is inflated into a total theory about the environment, the specific control that needs repair disappears. Effective change therefore depends on addressability - the ability to name a specific state error, control, procedure, or dependency that can be changed and re-tested.

9.2 Normalized dysfunction

Normalized dysfunction is a condition in which repeated failure becomes part of the accepted narrative rather than remaining visible as a correctable deviation. Statements such as “that department never responds,” “the software always crashes there,” or “that is just how the process works” may describe a real baseline, but they should not automatically convert malfunction into legitimacy.

Change principle

A baseline can describe what commonly happens without proving that what commonly happens is acceptable, efficient, safe, or aligned with the system’s stated purpose.

9.3 Correctability scale

Level	Name	Behavior after discrepancy is
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		demonstrated
CR0	Self-correcting	System detects and resolves the mismatch automatically.
CR1	Responsive	Correction occurs after a clear report or request.
CR2	Frictional	Correction is possible but requires repeated escalation or workaround.
CR3	Defensive	System primarily repeats or protects the presentation rather than repairing state.
CR4	Suppressive	Reporting or corrective mechanisms themselves are impaired, discouraged, or removed.

This scale is intended as design vocabulary rather than an established industry standard. It provides a way to test whether an environment can change after evidence is presented.

10. Failure Modes and Safeguards

Failure mode	Description	Safeguard
Presentation capture	The claim is accepted because an authority, interface, or local rendering says it.	Require state or mechanism validation for material claims.
Attribution inflation	Mismatch is immediately promoted into motive or coordination.	Evidence ladder; separate proposition grades.
Over-correlation	Temporal proximity is treated as causation.	Dependency, intervention, and severance tests.
Single-trial superstition	One inconvenience is treated as a stable pattern.	Replication under comparable conditions.
Perspective overreach	One camera, log, or person is treated as the complete environment.	Multi-view reconciliation and provenance.
Sensor monoculture	Several observations share the same hidden failure source.	Document source independence and common-mode dependencies.
Log completeness mistaken for	A complete-looking ledger is	Integrity checks, coverage

truth	assumed infallible.	analysis, alternate telemetry.
HUD contamination	Inference or prediction is written as historical fact.	Separate labels and immutable historical store.
Confirmation bias	Auditor selects only supporting evidence.	Competing hypotheses and disconfirmation tests.
Uncorrectable framing	Analysis cannot identify an addressable control or action.	Require a correction step or explicitly record why none is currently available.

11. Operational Templates

11.1 Full investigation record

Field	Record
Case / test ID	
Environment or system	
Date/time window	
Task or objective under review	
Presented claim or rendering	
Expected consequence if claim is correct	
Observed state or outcome	
Designed procedure	
Control surface	
Known identities/processes	
Relevant roles/permissions	
Independent records	
Candidate triggers	
Observed dependencies	
Replication result	
Intervention performed	
State change after intervention	
Competing explanations	

Evidence level - occurrence	
Evidence level - mechanism	
Evidence level - attribution	
Evidence level - causation	
Evidence level - coordination	
Evidence level - intent	
Prediction or inference labels	
What would disconfirm the leading explanation	
Unresolved questions	
Corrective action	
Correction response	
Re-test result	
Recurrence	
Reviewer / audit date	

11.2 Compact operating card

Prompt	Question
CLAIM	What was said, displayed, or implied?
STATE	What did the object, process, service, or environment actually do?
CONTROL	What mechanism changes that state?
PROCEDURE	What does the environment's own design require?
TRIAL	Does the mismatch reproduce?
PROVENANCE	Who or what produced the transition?
GRADE	What is recorded, derived, correlated, inferred, predicted, or unknown?
ALTERNATIVE	What competing explanation remains viable?
FIX	What specific action addresses the demonstrated condition?
RE-TEST	Did the environment change and remain

	changed?
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12. Proposed Research Program

12.1 Primary hypothesis

Primary hypothesis

Observers with access to provenance, demonstrated mechanisms, and controlled intervention will distinguish presentation-state discrepancies more accurately than observers relying primarily on rendered presentation, while explicit evidence grading will reduce unsupported causal and intentional attribution.

12.2 Suggested experimental groups

Group	Information available
P	Presentation only.
P + O	Presentation plus ordinary observations / telemetry.
P + PR	Presentation plus provenance and dependency information.
P + PR + I	Presentation plus provenance plus intervention controls.
P + PR + I + HUD	Full audit condition including labeled prediction / plan-recognition information.

Outcomes can be compared for discrepancy detection, causal accuracy, false positives, coordination errors, intent inflation, time-to-correction, and correction persistence.

12.3 Falsifiability requirements

- Every hypothesis must specify what observation would count against it.
- The framework must be capable of returning “presentation confirmed.”
- The framework must be capable of returning “unresolved.”
- Control conditions must include genuine randomness, ordinary mistakes, and emergent behavior.
- Intent must never be inferred solely from inconvenience, timing, annoyance, or pattern recognition.
- A missing causal link is an observability gap, not automatic evidence of concealment.
- The auditor itself must be scored for false narratives and selective evidence use.

13. Relevance to Systems Architecture and Virtual Environments

The framework intersects with established systems concepts without claiming that PSIF itself is an industry-standard term. Relevant areas include observability, distributed tracing, event sourcing, provenance, Byzantine behavior, equivocation, masquerading, anomaly detection, data fusion, causal inference, plan recognition, and audit logging. The source research documents identify NIST audit guidance, OpenTelemetry, W3C PROV, MITRE ATT&CK, and AI risk-management practices as useful reference anchors for these adjacent concepts.

What PSIF adds is an integrated research question: not merely “what happened?” or “was the telemetry correct?”, but “how faithfully does the environment’s presentation represent operational state, causal structure, and correctability?” This is particularly relevant in simulations where the rendered participant experience and the underlying engine state can be recorded simultaneously.

14. Relevance to Real and Hybrid Environments

The framework is equally valuable where no canonical engine state exists, provided its language is adjusted. Real-world analysis should use best-supported evidenced state rather than claims of universal ground truth. Hybrid environments - digital twins, corporate simulations, training systems, games, augmented reality, operational sandboxes, and environments containing real-world brands or procedures - may combine both forms of evidence.

The practical goal is not to maintain an artificial separation between “fictional” and “real” examples. The goal is to preserve the correct evidentiary standard in each setting. A simulation may expose full internal state. A real business interaction may expose only a control surface, documented procedure, repeated trial, and observable outcome. Both can still be analyzed using the same high-level sequence.

15. Final Synthesis

The concept began as a question about agents in a virtual environment attempting to rely on ordinary-looking environmental behavior and fragmented viewpoints to shape how activity is interpreted. It developed into a broader inquiry about the difference between representation and operation, and then into a practical concern about whether environments can identify and correct their own underlying problems.

The strongest independent formulation is therefore not “assume deception.” It is “preserve state-claim integrity.” When presentation and state diverge, identify the mechanism; when mechanisms are linked, reconstruct provenance; when causation is uncertain, intervene; when motive is uncertain, leave it uncertain; and when a problem is demonstrated, require an addressable correction and test whether the system changes.

Independent final formulation

Presentation-State Integrity is the degree to which an environment’s claims, renderings, narratives, and interfaces correspond to its evidenced operational state and causal structure. A state-claim divergence exists when those layers materially differ. The divergence should be reconstructed through observation, mechanism testing, provenance, replication, and

intervention before analysts evaluate whether its cause is error, structural failure, emergence, strategic behavior, or deliberate presentation-layer deception. The ultimate purpose is not merely to identify inconsistency but to expose an addressable condition, apply a correction, and determine whether the environment is capable of changing in response to demonstrated evidence.

Compact rule

Do not mistake presentation for state, and do not mistake suspicion for proof. Observe. Separate. Reproduce. Trace. Intervene. Compare. Grade. Correct. Re-test.

Appendix A. Source Materials Reviewed

- Presentation Versus Ground Truth: A testable framework for virtual and real-world environments. Multi-AI collaborative pass-through research document, September 3, 2026. Supplied in DOCX and PDF formats.
- Presentation vs State: Discernment Template. OAL working document, September 3, 2026. Draft for test design.
- User-provided original simulation prompt and subsequent clarification emphasizing discernment, correction, creativity, growth, and real-world testability.
- Prior ChatGPT independent development in the current collaborative pass-through, including the proposed Presentation-State Integrity Framework and related terminology.

The attached research documents characterize Systemic Presentation-Layer Deception and Provenance-Based Adversarial Observability as working design vocabulary rather than adopted standards. This document preserves that qualification and likewise treats PSIF, state-claim divergence, presentation capture, perspective overreach, presentation normalization, normalized dysfunction, and correction resistance as proposed design vocabulary unless otherwise noted.

Appendix B. Three-Outcome Decision Rule

Presentation confirmed: Available evidence supports the presented state/explanation at the required level. Next step: Close the case or continue routine monitoring.

Presentation contradicted: Material evidence establishes a state, mechanism, or causal relationship inconsistent with the presentation. Next step: Name the addressable condition, correct it, and re-test.

Unresolved: Available evidence cannot yet distinguish the competing explanations. Next step: Preserve uncertainty and identify the specific evidence or intervention needed next.